Algebraic structure with one binary operation :-

☆ Groupoid :-

A non empty set G with binary operation '*' is called groupoid if it satisfies the following property.

closure : $\forall a, b \in G \Rightarrow a * b \in G$.

e.g :- $(N, +)$, $(Z, +)$

☆ Semi group :-

A non empty set G with binary operation '*' is called semi group if '*' satis. the following properties.

closure : $\forall a, b \in G \Rightarrow a * b \in G$

Associative : $\forall a, b, c \in G$,

$$(a * b) * c = a * (b * c).$$

e.g :- $(N, +)$, $(I, +)$, $(N, \cdot)$

☆ Monoid :-

A non empty set G with binary operation '$*$' is called Monoid if '$*$' satisfies the following

Closure : $\forall a, b \in G \Rightarrow a * b \in G$.

Associative : $\forall a, b, c \in G$
$\Rightarrow (a * b) * c = a * (b * c)$.

Existence of Identity :

$\forall a \in G, \exists e \in G \ni a * e = e * a = a$.

e.g. : $(I, +)$ is a monoid ;
$(N, +)$ is not a monoid.

☆ Group :-

A non empty set G with binary operation '$*$' is called group, if '$*$' satisfies the following.

Closure :- $\forall a, b \in G \Rightarrow a * b \in G$

Associative :- $\forall a, b, c \in G \Rightarrow (a * b) * c = a * (b * c)$.

Existence of identity :- $\exists e \in G \ni a * e = e * a = a$

Existence of inverse :- $\forall a \in G, \exists b \in G \ni$
$a * b = e$.